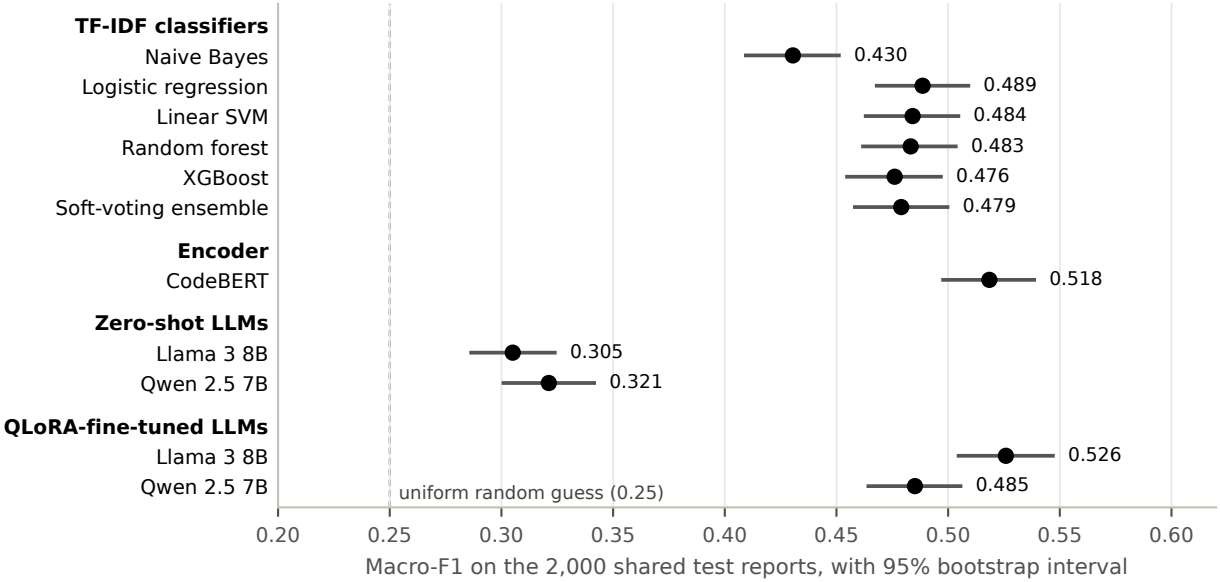
TF-IDF classifiers
Naive Bayes
0.430
Logistic regression
0.489
Linear SVM
0.484
Random forest
0.483
XGBoost
0.476
Soft-voting ensemble
0.479
Encoder
CodeBERT
0.518
Zero-shot LLMs
Llama 3 8B
0.305
Qwen 2.5 7B
0.321
QLoRA-fine-tuned LLMs
Llama 3 8B
0.526
Qwen 2.5 7B
0.485
uniform random guess (0.25)
0.20
0.25
0.30
0.35
0.40
0.45
0.50
0.55
0.60
Macro-F1 on the 2,000 shared test reports, with 95% bootstrap interval